

Ancestry-diagnostic loci are enriched in climate-associated regions, but directional sorting is not

Draft results summary (Module C, the extrinsic-climate arm). Methods are deferred to Supplementary Materials. Findings are descriptive: climate associations are genotype–environment correlations, and the sorting itself is licensed as selection only against the forthcoming neutral null. The identity of the climate axes (which variables load on PC1 versus PC2) is pending.

Results

Design. Per-population allele frequencies were associated with two per-population climate principal components (PC1, PC2) using BayPass across eight configurations: two population sets, two climate axes, and two covariance settings. Because sorting is by definition ancestry fixation, a climate axis that is itself correlated with genome-wide ancestry could co-localise with sorting for structural rather than ecological reasons. PC1 is only weakly ancestry-correlated across populations (Spearman $\rho = -0.23$); PC2 is moderately so ($\rho = +0.57$), driven predominantly by a single population, Åland. We therefore condition on the among-population covariance matrix estimated from the LD-pruned marker set, and take the Åland-excluded population set as the primary analysis, with the remaining configurations as comparators.

Measuring climate association per linkage cluster. We summarise each LD cluster’s climate association by its *significance rate*: the proportion of its member loci reaching $\text{BF}(\text{dB}) \geq 15$. A rate is used rather than a count of significant members because a count makes a cluster’s eligibility depend on its size—a cluster of nine loci (the median) would need 56% of its members significant to reach a threshold of five, whereas a cluster of ninety needs only 6%—so a count-based criterion applies a far stricter standard to small clusters and cannot be corrected by conditioning on size alone. All tests treat the cluster as the unit of observation, since member loci within a cluster are in linkage disequilibrium and are not independent; cluster size is retained as a covariate throughout. Because a rate is estimated imprecisely when a cluster contains few loci, results are reported across nested size strata.

Ancestry-diagnostic loci concentrate in climate-associated regions. Among clusters large enough for the significance rate to be well estimated (≥ 50 loci), the proportion of ancestry-diagnostic members ($\text{DI} > -25$) increases with climate association: each +10 percentage points of significance rate corresponds to +4.5 percentage points of diagnostic loci for PC1 ($p = 8 \times 10^{-4}$) and +3.7 for PC2 ($p = 0.013$), against a baseline of roughly 10%. A size-weighted, overdispersion-corrected model gives concordant estimates (odds ratios 1.50 and 1.47 per +10 percentage points). The effect is attenuated in the intermediate stratum (≥ 20 loci: +1.9 and +1.8 percentage points) and is not detectable when all clusters are pooled, as expected when the predictor is measured with substantial error in small clusters. Both climate axes behave alike, at similar magnitude.

Directional sorting shows no such association. Applying the same size-normalised, cluster-level test to directional ancestry sorting yields no relationship in the size-comparable stratum (odds ratios 0.99 for PC1 and 1.03 for PC2, $p > 0.95$); a member-level analysis is likewise null. This is the expected result on mechanistic grounds: genotype–environment association is detected through variation in allele frequency *among* populations, whereas unidirectional sorting drives most populations toward the same near-fixed state and therefore minimises exactly that variance.

Climate-driven selection would in any case predict populations diverging in different directions according to local conditions, rather than the parallel fixation that unidirectional sorting describes.

Interpretation and outlook. The climate signal in these data is a *static* one: regions whose allele frequencies track climate are enriched for ancestry-diagnostic loci, on both climate axes, by a modest but consistent margin. There is no corresponding *dynamic* signal—directional ancestry sorting is not preferentially located in climate-associated regions. We note the limits: the informative stratum comprises 878 clusters, the enrichment is a few percentage points against a variable background, and these are associations rather than demonstrations of selection. Identifying the climate axes and evaluating the sorting against a recombination-matched neutral null remain the decisive next steps; intrinsic incompatibilities are pursued separately through among-population linkage disequilibrium between unlinked diagnostic units.